

HARBOR BEACH, MI, USA

WMO: 997260

Lat: 43.850N

Lon: 82.640W

Elev: 179

StdP: 99.19

Time Zone: -5.00 (NAE)

Period: 04-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-17.0	-14.7	N/A	N/A	N/A	N/A	N/A	N/A	21.2	-7.0	16.8	-5.5	5.3	250	0.640

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	6.0	29.0	N/A	27.1	N/A	25.4	N/A	N/A	N/A	N/A	N/A	N/A	N/A	5.9	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 15.5	12.7	10.9	DB	-19.7	31.8	4.1	1.8	-22.7	33.1	-25.1	34.2	-27.4	35.2	-30.4	36.5	(4)
(5)			WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	7.9	-5.3	-5.0	-0.4	5.5	11.4	16.7	20.6	20.2	16.7	10.8	4.1	-0.8	(6)	
	DBStd	10.01	5.51	5.24	4.97	3.85	4.27	3.73	3.26	2.64	3.47	4.02	4.51	4.30	(7)	
	HDD10.0	1953	473	419	324	145	34	1	0	0	1	39	183	334	(8)	
	HDD18.3	3997	732	652	581	386	219	75	14	10	71	237	427	593	(9)	
	CDD10.0	1204	0	0	2	9	78	202	329	316	202	62	6	0	(10)	
	CDD18.3	206	0	0	0	0	4	26	85	67	22	2	0	0	(11)	
	CDH23.3	890	0	0	0	3	29	141	399	226	80	12	0	0	(12)	
	CDH26.7	205	0	0	0	0	4	34	101	46	17	3	0	0	(13)	
(14) Wind	WSAvg	5.2	5.8	5.6	5.8	6.0	4.9	4.7	4.2	4.3	4.4	5.2	5.1	5.9	(14)	
(15) Precipitation	PrecAvg	810	51	48	46	76	86	80	80	74	82	68	61	55	(15)	
	PrecMax	1021	94	140	115	141	167	158	138	122	188	140	143	104	(16)	
	PrecMin	614	14	13	17	22	43	26	24	16	11	40	17	23	(17)	
	PrecStd	94	22	29	25	32	37	38	27	26	39	27	30	22	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	10.8	10.0	16.7	21.9	27.2	30.3	32.0	30.5	28.8	24.5	16.9	11.7	(19)	
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(20)	
	2%	DB	6.9	5.7	12.4	16.8	23.6	27.0	29.3	27.7	25.7	20.5	14.1	8.6	(21)	
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(22)	
	5%	DB	3.9	3.1	8.0	13.2	21.0	24.7	27.4	25.7	23.2	18.1	12.2	6.4	(23)	
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(24)	
	10%	DB	2.0	1.3	5.5	10.5	18.5	22.3	25.5	24.1	21.7	16.4	10.4	4.6	(25)	
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(27)	
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(28)	
	2%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(29)	
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(30)	
	5%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(31)	
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(32)	
	10%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(33)	
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	5.9	6.0	5.8	6.0	7.1	6.3	6.0	5.7	5.9	5.7	5.1	4.7	(35)	
		MCDBR	8.1	7.7	10.2	11.8	12.0	10.4	9.2	8.1	8.4	8.3	8.0	7.0	(36)	
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(37)	
	5% WB	MCDBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(38)	
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(39)	
(40) Clear-Sky Solar Irradiance	taub		0.306	0.307	0.316	0.362	0.386	0.414	0.411	0.402	0.373	0.356	0.318	0.287	(40)	
	taud		2.392	2.365	2.413	2.345	2.351	2.297	2.340	2.352	2.423	2.438	2.509	2.508	(41)	
	Ebn at Noon		818	886	925	901	885	858	856	852	853	819	805	817	(42)	
	Edh at Noon		77	93	101	118	121	129	122	117	100	87	69	63	(43)	
(44) All-Sky Solar Radiation	RadAvg		1.19	2.02	3.43	4.55	5.38	6.05	6.02	5.18	4.09	2.51	1.44	0.93	(44)	
	RadStd		0.13	0.22	0.35	0.40	0.47	0.35	0.28	0.24	0.38	0.25	0.20	0.11	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (43 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page